Lecture 17: multimodal models

> 第 17 讲：多模态模型

So far: language models

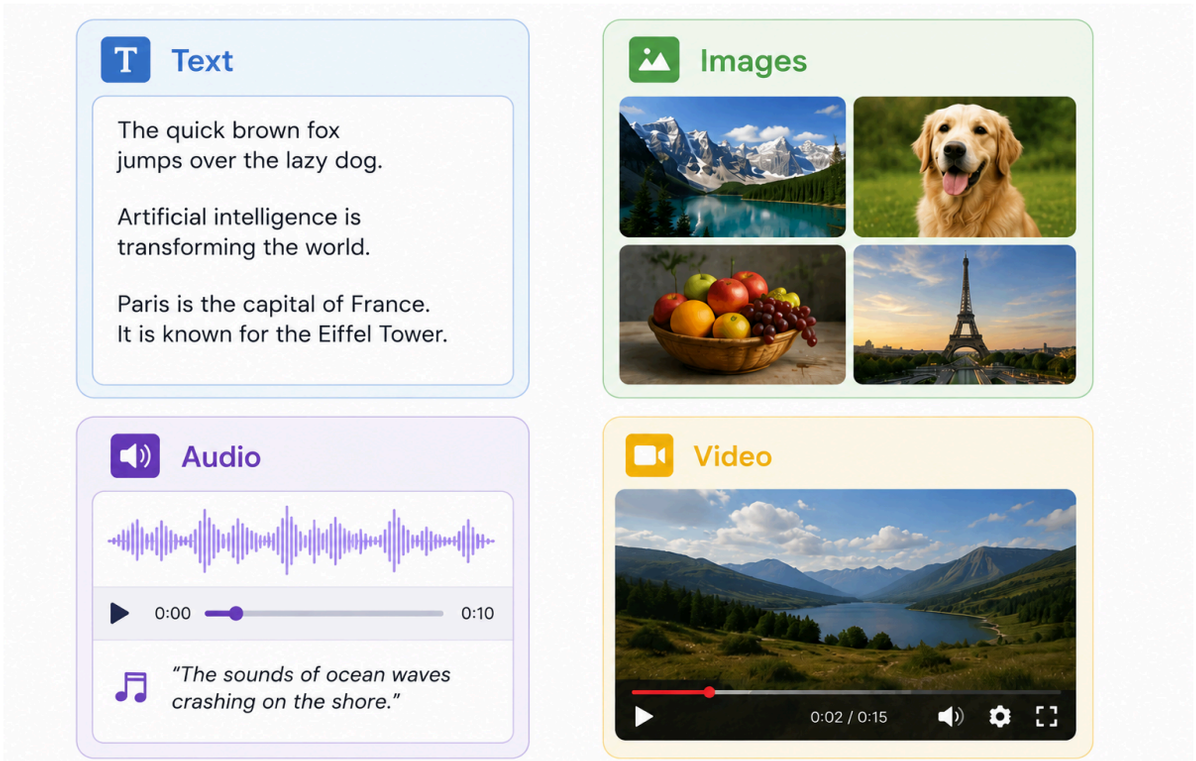
> 目前为止：语言模型

text ⇒ text

> 文本 ⇒ 文本

The world is multimodal:

> 世界是多模态的：



Ultimate goal: **omni model**

> 终极目标：omni 模型

- Input any combination of modalities (understanding)

> 输入任意模态组合（理解）

- Output any combination of modalities (generation)

> 输出任意模态组合（生成）

Where we are today:

> 我们今天的现状：

- Transformers work really well. So we gotta use them.

> Transformer 效果非常好，所以我们得用它。

- Transformers speak tokens (discrete or continuous), where a token represents some ~semantic unit of information.

> Transformer 以 token 为语言（离散或连续），每个 token 大致代表一个语义层面的信息单元。

- Therefore, we must convert everything into tokens.

> 因此，我们必须把一切都转换成 token。

- Note: we had to do this with text (recall the tokenization lecture).

> 注意：对文本我们也不得不这么做（回忆 tokenization 那一讲）。

- For non-text modalities, this is more challenging...

> 对非文本模态来说，这更具挑战性……

Questions:

> 问题：

1. How do we input non-text data (e.g., understand images)?

> 如何输入非文本数据（例如理解图像）？

2. How do we output non-text data (e.g., generate audio)?

> 如何输出非文本数据（例如生成音频）？

Summary:

> 总结：

- Frontier models are expected to be multimodal (natively multimodal, omni)

> 前沿模型被期望是多模态的（原生多模态、omni）

- Fundamental challenge: how to encode non-text modalities?

> 根本挑战：如何编码非文本模态?

- Comprehension and generation might demand different things (semantics versus finer-grained details)

> 理解与生成可能需要不同的东西（语义 vs 更细粒度的细节）

- Balance images + video (lower information density) and text for training stability

> 为了训练稳定性，需要在图像 + 视频（信息密度较低）与文本之间取得平衡

- Continuous encoders + Transformer + diffusion models for generation

> 连续编码器 + Transformer + 扩散模型负责生成

CLIP (Contrastive Language-Image Pretraining) [Radford+ 2021]

> CLIP（对比式语言-图像预训练） [Radford+ 2021]

Context:

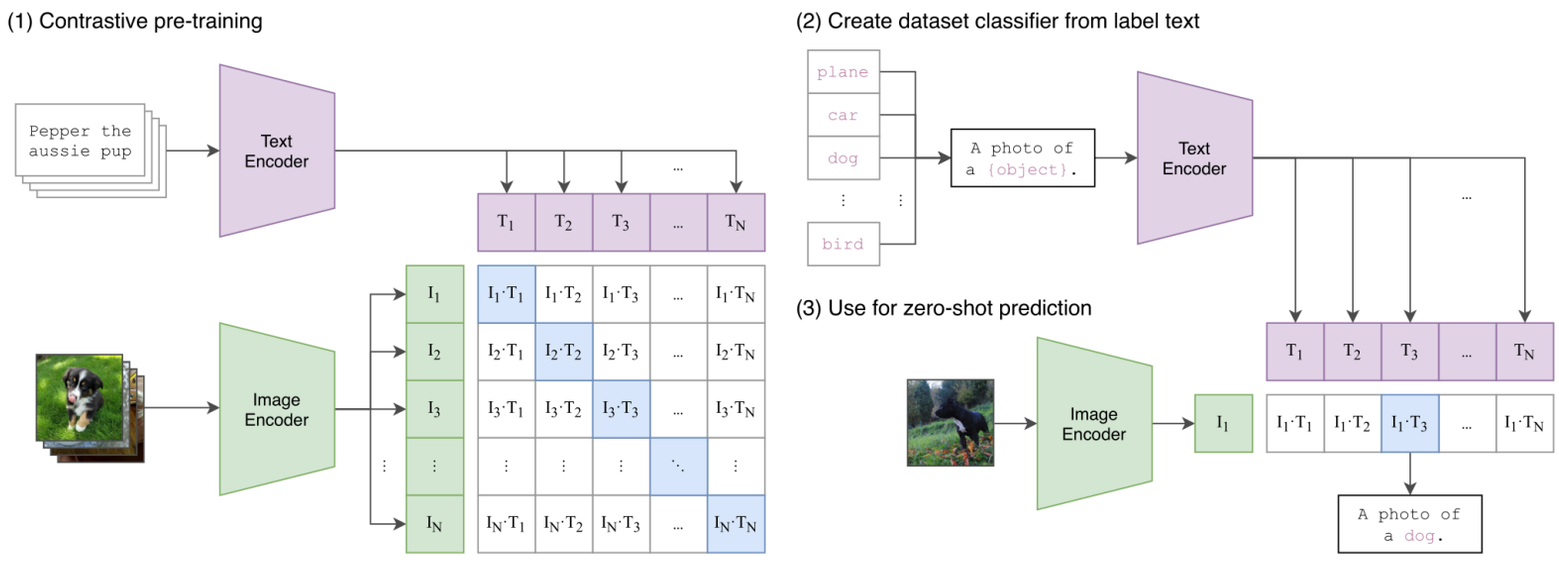
> 背景：

- Computer vision models were trained on annotated images.

> 计算机视觉模型是在带标注的图像上训练的。

- Question: is it possible to leverage the much larger amount of (image, caption) pairs?

> 问题：能否利用规模大得多的（图像, 描述）对?



Method:

> 方法：

- Get a batch of (image, text) examples (e.g., 32768)

> 取一批（图像, 文本）样本（例如 32768 个）

- Encode each image and each text

> 编码每张图像和每段文本

- For each image, prefer its aligned text over other texts

> 对每张图像，偏向其配对的文本而非其他文本

- For each text, prefer its aligned image over other images

> 对每段文本，偏向其配对的图像而非其他图像

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t             - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

Data:

> 数据：

- Searched for 500K queries, get ~20K (image, text) pairs per query

> 搜索了 500K 个查询，每个查询得到约 20K 个（图像, 文本）对

- Trained on 400M image-text pairs

> 在 400M 图文对上训练

- Didn't release the dataset

> 没有公开数据集

- Reproduced in OpenCLIP (using LAION-5B dataset, which used CLIP for filtering) [Cherti+ 2022]

> 在 OpenCLIP 中被复现（使用 LAION-5B 数据集，而该数据集本身用 CLIP 做了过滤） [Cherti+ 2022]

Data processing [code]

> 数据处理 [code]

- Images come in all resolutions (arbitrary W x H)

> 图像有各种分辨率（任意的 W x H）

- Resize using bicubic interpolation so shorter side is 336 pixels

> 用双三次插值调整大小，使较短边为 336 像素

- Center crop (cuts off borders to get 336 x 336)

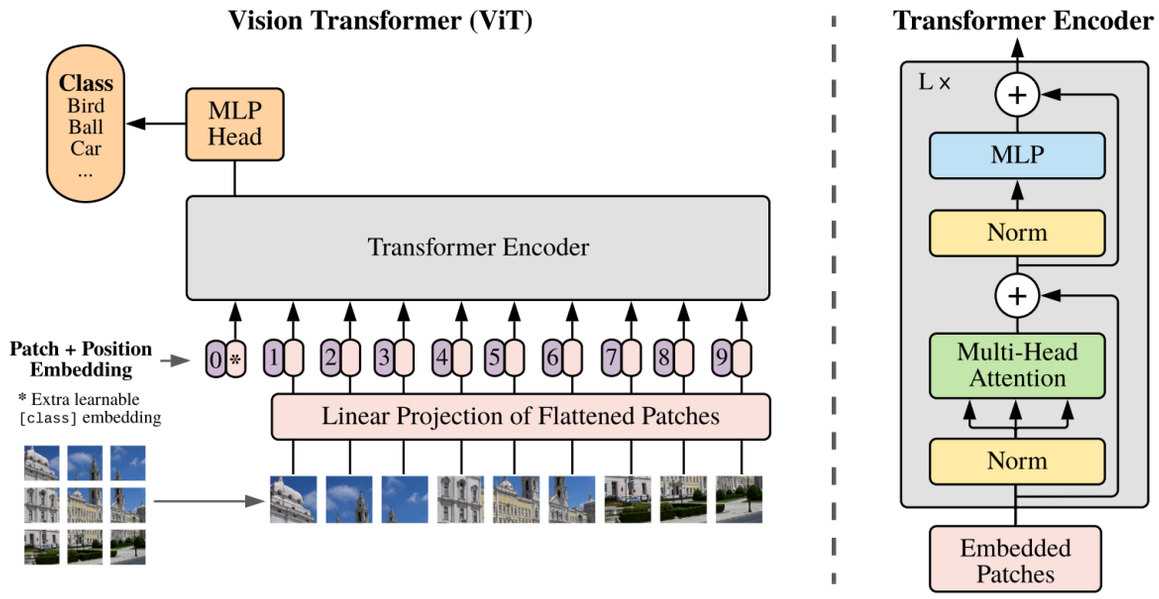
> 中心裁剪（切掉边缘以得到 336 x 336）

Vision encoder:

> 视觉编码器：

- Experimented with ResNet-50 and Vision Transformers [Dosovitskiy+ 2020]

> 实验了 ResNet-50 和 Vision Transformer [Dosovitskiy+ 2020]



- Attention pooling: do QKV with query = global average of activations

> 注意力池化（attention pooling）：以 query = 激活值的全局平均来做 QKV

- Best model: ViT-L/14@336px (L = large, 14x14 patches, 3 channels, trained on 336x336 resolution images)

> 最佳模型：ViT-L/14@336px（L = large，14x14 patch，3 通道，在 336x336 分辨率图像上训练）

Text encoder:

> 文本编码器：

- GPT-2 Transformer (63M parameters, 12 layers)

> GPT-2 Transformer（63M 参数，12 层）

- Encode [BOS] ... [EOS], return [EOS] activation at highest layer

> 编码 [BOS] ... [EOS]，返回最高层的 [EOS] 激活值

Headline result:

> 主要结果：

- On ImageNet, zero-shot CLIP outperformed ResNet-50 trained on 1.2M ImageNet images

> 在 ImageNet 上，zero-shot CLIP 超过了在 1.2M 张 ImageNet 图像上训练的 ResNet-50

Ablation:

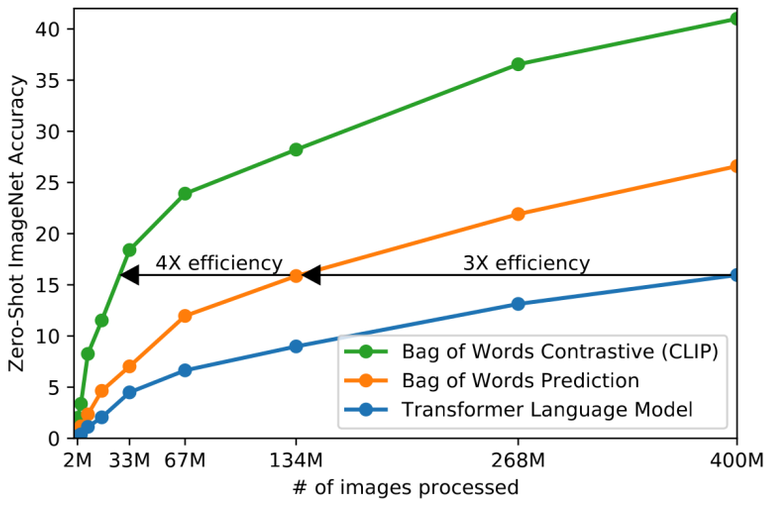
> 消融实验：

- Alternative: predict text from images directly

> 替代方案：直接从图像预测文本

- Much less compute efficient compared to CLIP-style ranking

> 与 CLIP 式排序相比，计算效率低得多



Summary:

> 总结：

- Encoding of images captures semantics given by (noisy) text

> 图像的编码捕捉到了（带噪声的）文本所给出的语义

- Design decisions chosen based on image classification (not very fine-grained)

> 设计决策是基于图像分类任务做出的（并不很细粒度）

- Technical: requires large batch sizes, softmax operation over full batch

> 技术层面：需要很大的 batch size，且 softmax 要在整个 batch 上做

SigLIP (Sigmoid Loss for Language Image Pre-Training) [Zhai+ 2023]

> SigLIP（用于语言-图像预训练的 Sigmoid 损失） [Zhai+ 2023]

Objective:

> 目标函数：

- CLIP: multiclass classification for (text, image) versus (text, image') for all image'

> CLIP：将 (文本, 图像) 与所有 (文本, 图像') 做多分类

- SigLIP: binary classification for (text, image) - aligned or not?

> SigLIP：对 (文本, 图像) 做二分类——是否配对?

Algorithm 1 Sigmoid loss pseudo-implementation.

```
# img_emb      : image model embedding [n, dim]
# txt_emb      : text model embedding [n, dim]
# t_prime, b   : learnable temperature and bias
# n            : mini-batch size

t = exp(t_prime)
zimg = l2_normalize(img_emb)
ztxt = l2_normalize(txt_emb)
logits = dot(zimg, ztxt.T) * t + b
labels = 2 * eye(n) - ones(n) # -1 with diagonal 1
l = -sum(log_sigmoid(labels * logits)) / n
```

Data:

> 数据：

- WebLI dataset: O(billion) (image, text) pairs [Chen+ 2022]

> WebLI 数据集：O(billion) 量级的（图像, 文本）对 [Chen+ 2022]

- Scraped from the Internet

> 从互联网抓取

- Used automatic OCR to extract text from images

> 用自动 OCR 从图像中提取文本

- Keep 10% highest quality

> 保留质量最高的 10%

- Supports 100 languages

> 支持 100 种语言

Efficiency:

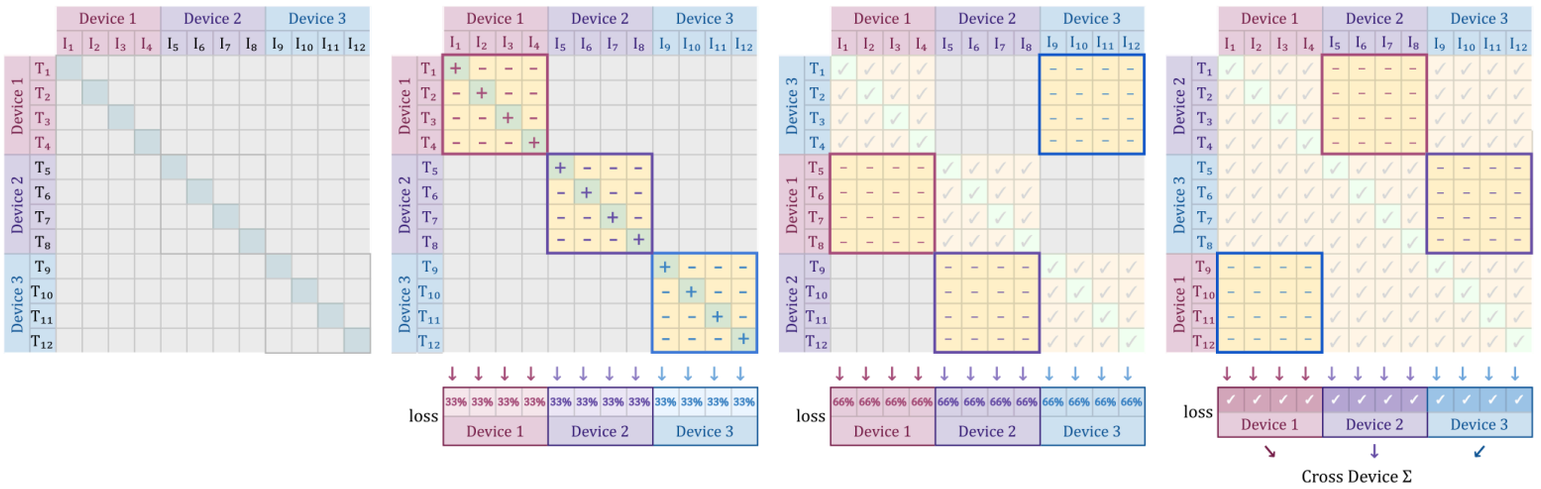
> 效率：

- CLIP: 10 days on 256 TPUv3

> CLIP：在 256 个 TPUv3 上 10 天

- SigLIP: 5 days on 32 TPUv4 (lower FLOP/s than TPUv3) - much faster!

> SigLIP：在 32 个 TPUv4 上 5 天（FLOP/s 低于 TPUv3）——快得多！



Batch size:

> Batch size：

- Decouple batch size from loss

> 将 batch size 与损失解耦

- Better than CLIP for <16K batch sizes

> 在 <16K 的 batch size 下优于 CLIP

- Go up to 1M batch size, but 32K is enough

> 可以上到 1M 的 batch size，但 32K 就够了

LLaVA (Large Language and Vision Assistant) [Liu+ 2023]

> LLaVA（大型语言与视觉助手） [Liu+ 2023]

Vision encoder: CLIP

> 视觉编码器：CLIP

Text decoder: Vicuna (LLaMA fine-tuned on ShareGPT conversations) [post]

> 文本解码器：Vicuna（在 ShareGPT 对话上微调的 LLaMA） [post]

Data:

> 数据：

- MS COCO has images annotated with bounding boxes and Mechanical Turk captions

> MS COCO 的图像带有边界框标注和 Mechanical Turk 撰写的描述

- Prompt GPT-4 with captions or detected objects and generate questions or conversations

> 用图像描述或检测到的物体去 prompt GPT-4，生成问题或对话

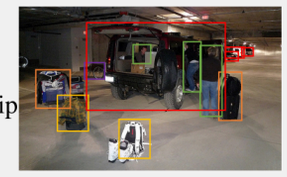
- Pair generations with original images

> 把生成结果与原始图像配对

- 158K examples

> 158K 个样本

Context type 1: Captions
A group of people standing outside of a black vehicle with various luggage.
Luggage surrounds a vehicle in an underground parking area
People try to fit all of their luggage in an SUV.
The sport utility vehicle is parked in the public garage, being packed for a trip
Some people with luggage near a van that is transporting it.



Context type 2: Boxes
person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation
Question: What type of vehicle is featured in the image?
Answer: The image features a black sport utility vehicle (SUV) ...<omitted>
Response type 2: detailed description
The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>
Response type 3: complex reasoning
Question: What challenges do these people face?
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

Model:

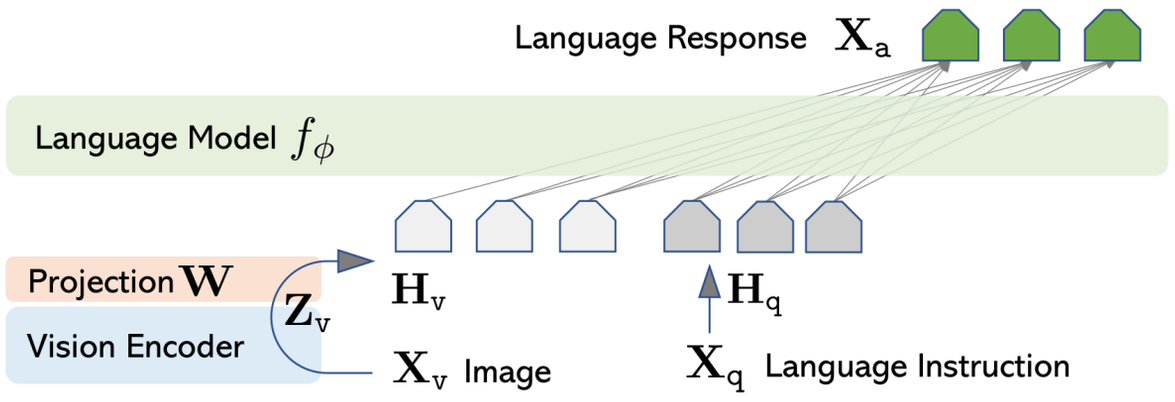
> 模型：

- Encode images with CLIP (ViT-L/14)

> 用 CLIP（ViT-L/14）编码图像

- Linear projection (W) into embedding space (Flamingo and Q-former are more complex)

> 线性投影（W）映射到嵌入空间（Flamingo 和 Q-former 更复杂）



Training:

> 训练：

- Stage 1 (alignment): freeze vision encoder and language model, only train W

> 阶段 1（对齐）：冻结视觉编码器和语言模型，只训练 W

- Stage 2 (fine-tuning): freeze vision encoder and train W and language model

> 阶段 2（微调）：冻结视觉编码器，训练 W 和语言模型

Visual input example, Extreme Ironing:



Source: https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg

User	What is unusual about this image?
LLaVA	The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment. [*Start a new conversation, and clear the history*]
User	What's happening in the scene?
LLaVA	The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention due to his unconventional choice of ironing his clothes on top of a moving car. The city street around him is bustling with activity, adding to the unique nature of the scene.
User	What is unusual about this image?
GPT-4 [36]	The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.
User	What is unusual about this image?
BLIP-2	a man is sitting on the back of a yellow cab
User	What is unusual about this image?
OpenFlamingo	The man is drying his clothes on the hood of his car.

LLaVA OneVision [Li+ 2024]

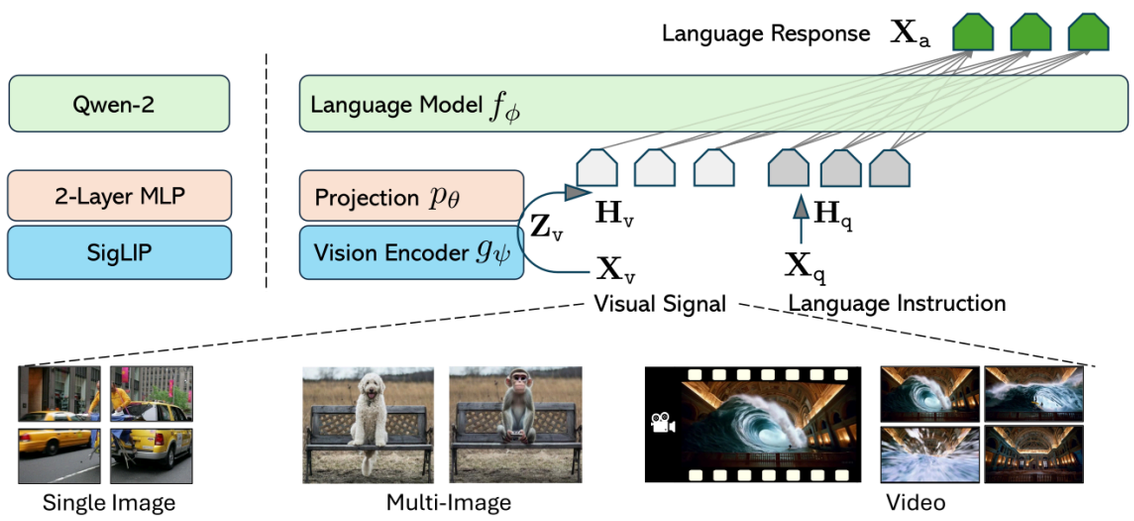
> LLaVA OneVision [Li+ 2024]

- Latest version in the LLaVA series (after LLaVA 1.5, LLaVA-Next)

> LLaVA 系列的最新版本（在 LLaVA 1.5、LLaVA-Next 之后）

- Handle multiple images, video

> 处理多图像、视频



- Vision encoder: SigLIP (use grid features before and after last Transformer layer)

> 视觉编码器：SigLIP（使用最后一个 Transformer 层前后的网格特征）

- Text decoder: Qwen-2 72B

> 文本解码器：Qwen-2 72B

- Projector: 2-layer MLP

> 投影器：2 层 MLP

Data processing:

> 数据处理：

- Preserving high resolution is important (e.g., for OCR)

> 保留高分辨率很重要（例如对 OCR）

- CLIP resizes and crops to 336x336, which loses information

> CLIP 会缩放并裁剪到 336x336，这会损失信息

- Solution: AnyRes, introduced in LLaVA 1.5 [paper]

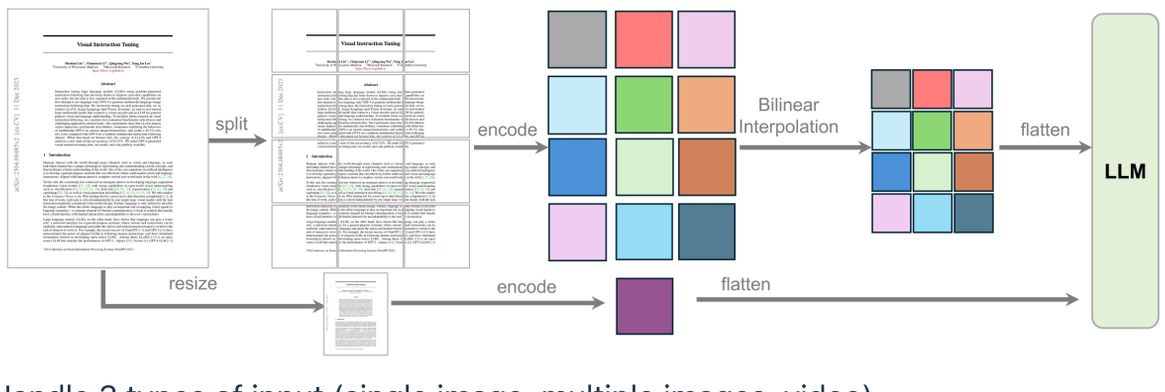
> 解决方案：AnyRes，在 LLaVA 1.5 中引入 [paper]

- Break up image into a x b pieces (matching resolution of vision encoder), encode, concatenate

> 把图像拆成 a x b 块（与视觉编码器的分辨率匹配），逐块编码，再拼接

- If too many tokens (original image is too high resolution), then use bilinear interpolation

> 如果 token 数太多（原图分辨率太高），就用双线性插值



Handle 3 types of input (single image, multiple images, video):

> 处理 3 类输入（单图像、多图像、视频）：

- Goal: make all of the modalities produce roughly the same length

> 目标：让所有模态产出的长度大致相同



- Single image: use higher resolution

> 单图像：用更高分辨率

- Multiple images: use base resolution for each image

> 多图像：每张图用基础分辨率

- Video: use lower resolution for each frame

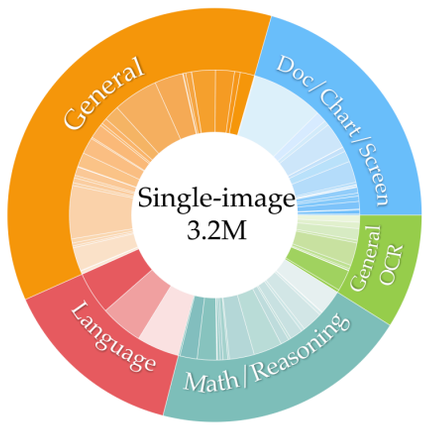
> 视频：每一帧用更低分辨率

Data:

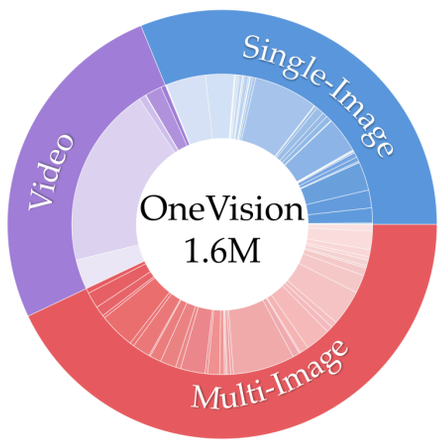
> 数据：

- Philosophy: quality over quantity

> 理念：质量重于数量



General (36.1%)	ALLaVA Inst (70.0K) [16]	AOKVQA (66.2 K)	Cambrian (filtered) (83.1 K)
CLEVR (0.7 K)	COCO Caption (20.0 K)	Hateful Memes (8.5 K)	IconQA (2.5 K)
Image Textualization (99.6 K)	LLaVA-158K (158.0 K)	LLaVA-Wild (train) (54.5 K)	LLaVAR (20.0 K)
OKVQA (9.0 K)	RefCOCO (50.6 K)	ScienceQA (5.0 K)	ShareGPT4o (57.3 K)
ShareGPT4V (91.0 K)	ST-VQA (17.2 K)	TallyQA (9.9 K)	Vision FLAN (186.1 K)
Visual7W (14.4 K)	VisText (10.0 K)	VizWiz (6.6 K)	VQARAD (0.3 K)
VQAv2 (82.8 K)	VSR (2.2 K)	WebSight (10.0 K)	InterGPS (1.3 K)
Doc/Chart/Screen (20.6%)	AI2D (GPT4V) (4.9 K)	AI2D (InternVL) (12.4 K)	AI2D (Original) (3.2 K)
Chart2Text (27.0 K)	ChartQA (18.3 K)	Diagram Image2Text (0.3 K)	Doc-VQA (10.2 K)
DVQA (20.0 K)	FigureQA (1.0 K)	HiTab (2.5 K)	Infographic VQA (4.4 K)
LRV Chart (1.8 K)	RoBUT SQA (8.5 K)	RoBUT WikiSQL (75.0 K)	RoBUT WTQ (38.2 K)
Screen2Words (15.7 K)	TQA (1.4 K)	UReader Caption (91.4 K)	UReader IE (17.3 K)
UReader KG (37.6 K)	UReader QA (252.9 K)	VisualMRC (3.0 K)	
Math/Reasoning (20.1%)	MAVIS MCollect (87.4 K)	MAVIS Data Engine (100.0 K)	Geo170K QA (67.8 K)
Geometry3K (2.1 K)	GEOS (0.5 K)	Geometry3K (MathV360K) (9.7 K)	GeoMVerse (MathV360K) (9.3 K)
GeoQA+ (MathV360K) (17.2 K)	MapQA (MathV360K) (5.2 K)	CLEVR-Math (5.3 K)	Geo170K Align (60.3 K)
MathQA (29.8 K)	Super-CLEVR (8.7 K)	TabMWP (45.2 K)	UniGeo (12.0 K)
GQA (72.1 K)	LRV Normal (10.5 K)	RAVEN (2.1 K)	Visual Genome (86.4K)
General OCR (8.9%)	ChromeWriting (8.8 K)	HME100K (74.5 K)	IIIT5K (2.0 K)
IAM (5.7 K)	K12 Printing (12.8 K)	OCR-VQA (80.0 K)	Rendered Text (10.0 K)
SynthDog-EN (40.1 K)	TextCaps (21.9 K)	TextOCR (25.1 K)	
Language (14.3%)	Magpie Pro (L3 MT) (150.0 K)	Magpie Pro (L3 ST) (150.0 K)	Magpie Pro (Qwen2 ST) (150.0 K)



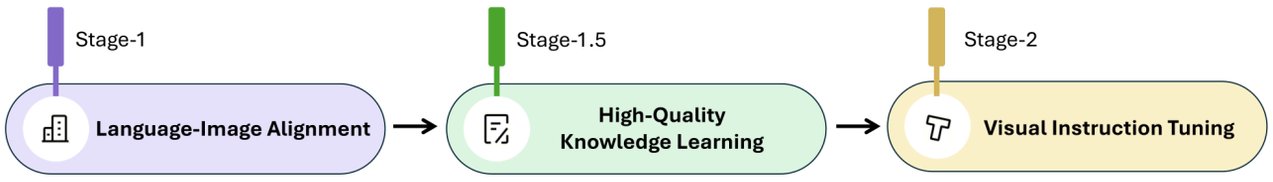
Single-Image (31.2%)	Magpie Pro (90.0K)	Vision FLAN (filtered) (55.8K)	Image Textualization (49.8K)
Cauldron (40.2K)	UReader (39.9K)	ShareGPT4V (21.0K)	ALLaVA Inst. (21.0K)
Cambrian (filtered GPT4o) (24.9K)	LLAVA-Wild (train) (10.9K)	LAION-GPT4V (8.0K)	LLAVA-158K (7.0K)
Geo170K-QA (6.8K)	Geo170K-Align (6.0K)	ShareGPT4o (5.7K)	TabMWP (4.5K)
LLAVAR GPT4 (4.0K)	MapQA (4.3K)	MathQA (3.0K)	TextOCR (GPT4V) (2.5K)
TextCaps (2.2K)	ScienceQA (1.9K)	FigureQA (1.8K)	GeoQA+ (1.7K)
AI2D (InternVL) (1.2K)	UniGeo (1.2K)	IconQA (1.1K)	LRV-Normal (filtered) (1.1K)
TQA (1.0K)	Geometry3K (1.0K)	Super-CLEVR (0.9K)	AI2D (GPT4V) (0.7K)
VizWiz (0.7K)	VQA-AS (0.6K)	CLEVR-Math (0.5K)	PlotQA (0.5K)
GEOS (0.5K)	InfoVQA (0.9K)	PMC-VQA (0.4K)	Geo3K (0.2K)
VQA-RAD (0.2K)	LRV-Chart (0.2K)		
Multi-Image (43.0%)	NLVR (86.4K)	Co-Instruct (50.0K)	ScanNet (49.9K)
RAVEN (35.0K)	IconQA (34.6K)	VIST (26.0K)	ScanQA (25.6K)
ContrastiveCaption (25.2K)	ALFRED (22.6K)	FlintstonesSV (22.3K)	ImageCode (16.6K)
DreamSim (15.9K)	Birds-to-Words (14.3K)	PororoSV (12.3K)	Spot-the-Diff (10.8K)
nuScenes (9.8K)	VISION (9.9K)	WebQA (9.3K)	RecipeQA-VisualCloze (8.7K)
RecipeQA-ImageCoherence (8.7K)	TQA (MI) (8.2K)	AESOP (6.9K)	HQ-Edit-Diff (7.0K)
MagicBrush-Diff (6.7K)	COMICS-Dialogue (5.9K)	MultiVQA (5.0K)	VizWiz (MI) (4.9K)
CLEVR-Change (3.9K)	NextQA (3.9K)	IEdit (3.5K)	Star (3.0K)
DocVQA (MI) (1.9K)	MIT-PropertyCoherence (1.9K)	MIT-StateCoherence (1.9K)	OCR-VQA (MI) (1.9K)
Video (25.9%)	ActivityNet (6.5K)	Charades (23.6K)	Ego4D (0.8K)
NextQA (9.5K)	ShareGPT4Video (255.0K)	Youcook2 (41.9K)	

Training:

> 训练：

- Philosophy: easier to harder

> 理念：由易到难



		Stage-1	Stage-1.5	Stage-2	
				Single-Image	OneVision
Vision	**Resolution**	384	384×{2×2, 1×{2,3}, {2,3}×1}	384×{{1×1}, ⋯, {6×6}}	384×{{1×1}, ⋯, {6×6}}
	#Tokens	729	Max 729×5	Max 729×10	Max 729×10 (See Fig. 3)
Data	**Dataset**	LCS	Image (Sec. 4.1)	Image (Sec. 4.2)	(Multi)-Image & Video (Sec. 4.2)
	#Samples	558K	4M	3.2M	1.6M
Model	**Trainable**	Projector	Full Model	Full Model	Full Model
	0.5B LLM	1.8M	0.8B	0.8B	0.8B
	7.6B LLM	20.0M	8.0B	8.0B	8.0B
	72.7B LLM	72.0M	73.2B	73.2B	73.2B
Training	**Batch Size**	512	256/512	256/512	256/512
	LR: ψ_{vision}	1×10^{-3}	2×10^{-6}	2×10^{-6}	2×10^{-6}
	LR: $\{\theta_{\text{proj}}, \phi_{\text{LLM}}\}$	1×10^{-3}	1×10^{-5}	1×10^{-5}	1×10^{-5}
	Epoch	1	1	1	1

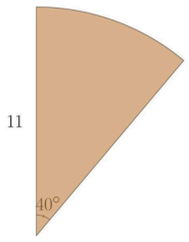
Transfer between modalities:

> 模态间的迁移：

- Single image data for diagrams and charts, but generalize to multi-image

> 示意图与图表任务用单图像数据，但能泛化到多图像

S1: Joint Understanding of Diagram and Chart from Multi-Image



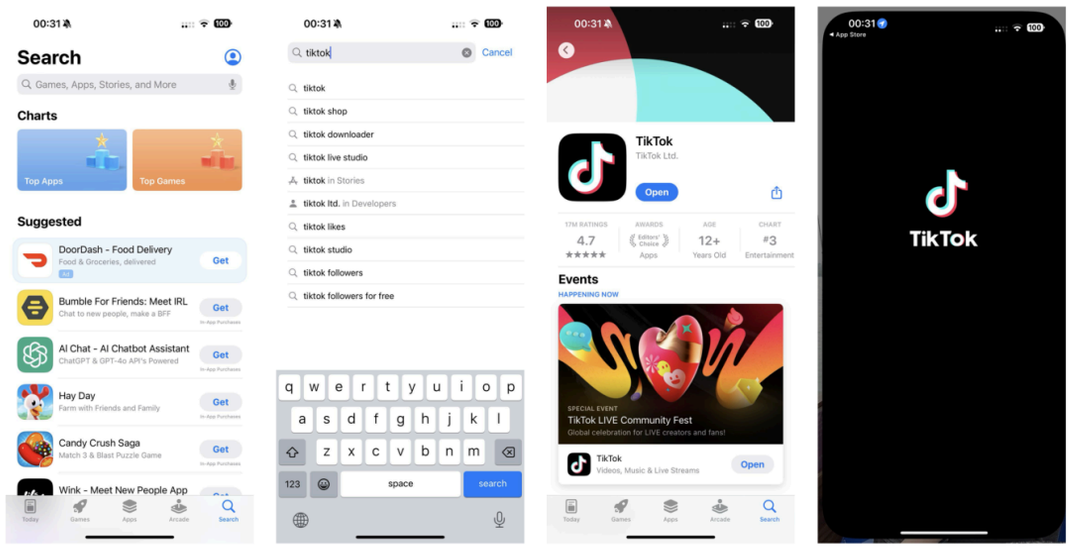
Insurance Company	Price
State Farm	68
Allstate	63
Liberty Mutual	59
USAA	90

User	Ross owns a house similar to the brown sector in the following image . They want to get the insurance from Allstate. The price per unit area is given in the following figure . What would be the cost of insuring the whole house? All the computations are rounded to two places of decimal.
LLaVA-OV	To find the cost of insuring the whole house, we need to calculate the area of the sector and then multiply it by the price per unit area. The area of the sector can be calculated using the formula $A = (\theta/360) * \pi * r^2$, where θ is the central angle and r is the radius. In this case, $\theta = 40°$ and $r = 11$. Plugging these values into the formula, we get $A = (40/360) * \pi * 11^2 \sim 38.01$. The price per unit area is given in the table as \$63. Multiplying the area by the price per unit area, we get the cost of insuring the whole house as $38.01 * \$63 \sim \$2,386.03$.

- OCR on single image data, relational reasoning from multi-image data, generalize to GUI-based agents

> OCR 用单图像数据，关系推理用多图像数据，泛化到基于 GUI 的 agent

S2: Multi-image GUI Understanding for Agents

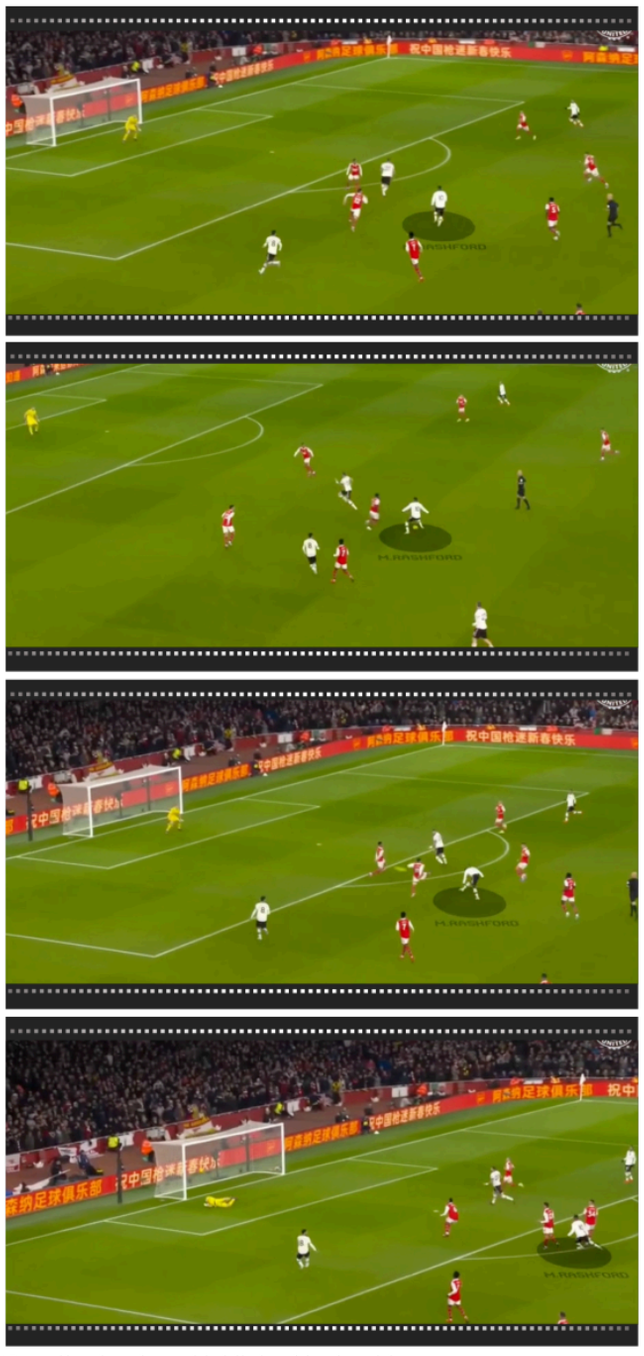


User	There are four sequential mobile UI screens, and are transformed by three steps of tap operations. Please respectively describe the three tap operations.
LLaVA-OV	The image shows a sequence of four mobile application interface screens. Here's a breakdown of the three tap operations based on the visible changes: 1. First Tap Operation: - The user starts by tapping on the search bar at the top of the screen. - The keyboard is shown at the bottom of the screen, indicating that the user can now type a search query. - The user types "TikTok" in the search bar and taps the search icon on the keyboard to initiate a search. 2. Second Tap Operation: - After tapping the search icon or pressing 'Enter', the screen displays a list of search results. - The user taps on the first result, which is the TikTok application. - This action opens the TikTok's app store page, which includes the app icon, developer information ... 3. Third Tap Operation: - The user taps the "Open" button on the app store page. - This action opens the TikTok application, and the screen transitions to the app's main interface.

- Visual prompting (circle) in single images, generalize to videos

> 在单图像中做视觉提示（画圈），泛化到视频

S8: Visual Prompting in Video



User	Describe the player highlighted in the video.
LLaVA-OV	The player highlighted in the video is wearing a white kit, and the number "10" is visible on their shirt. This player is positioned in the center of the pitch and is actively involved in the play, demonstrating control and dribbling skills as they navigate around the field. The player is surrounded by opponents in red kits, indicating that they are likely a key player for their team, possibly a playmaker or attacking midfielder. Throughout the video, the player maintains possession of the ball and creates opportunities for their team, showcasing their importance in the game's dynamics.

Summary:

总结：

- Standard VLM template: vision encoder + projector + LM

标准 VLM 模板：视觉编码器 + 投影器 + LM

- Most work goes into data curation (heavy on synthesized, task-specific data)

大部分功夫花在数据整理上（大量使用合成的、面向特定任务的数据）

- Open-source (released model weights and data)

开源（发布了模型权重和数据）

Qwen-VL [Bai+ 2023]

Qwen-VL [Bai+ 2023]

Architecture:

架构：

- Vision encoder: OpenCLIP's ViT-bigC (14x14 patches) [Cherti+ 2022]

视觉编码器：OpenCLIP 的 ViT-bigC（14x14 patch） [Cherti+ 2022]

- Adaptor: one layer cross-attention, incorporate 2D positional encodings, maps to fixed length of 256

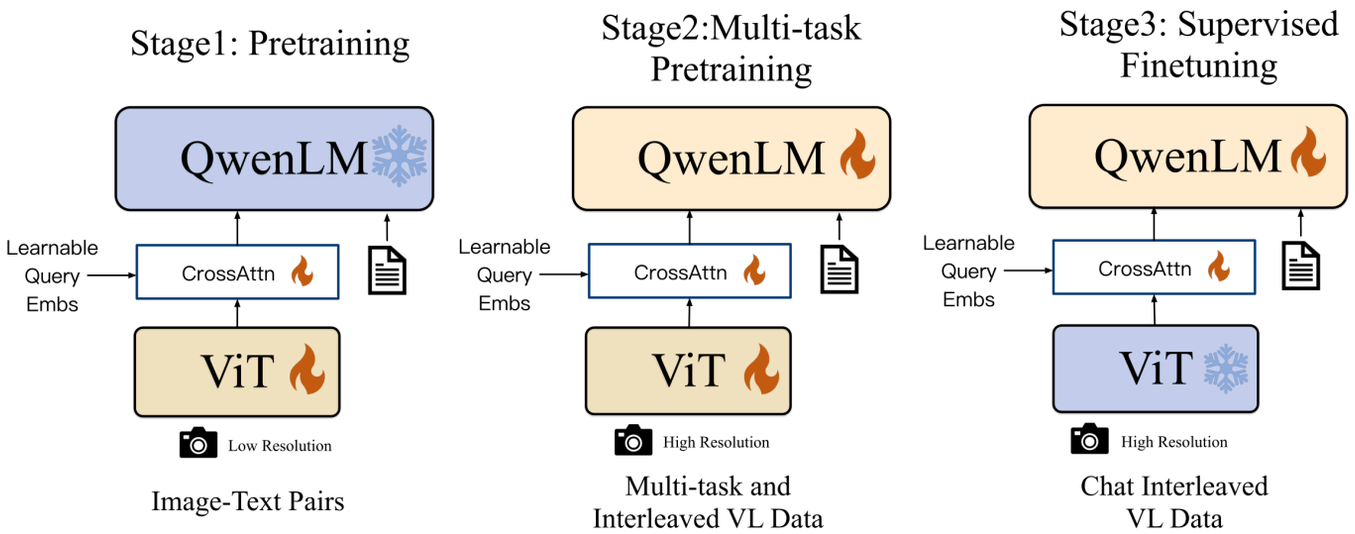
> 适配器：一层 cross-attention，融入 2D 位置编码，映射到固定长度 256

- Special tokens: , ,

> 特殊 token：, ,

Training:

> 训练：



- Stage 1: large-scale low quality data; freeze LM, train vision encoder + adaptor

> 阶段 1：大规模低质量数据；冻结 LM，训练视觉编码器 + 适配器

Language	Dataset	Original	Cleaned	Remaining%
English	LAION-en	2B	280M	14%
	LAION-COCO	600M	300M	50%
	DataComp	1.4B	300M	21%
	Coyo	700M	200M	28%
	CC12M	12M	8M	66%
	CC3M	3M	3M	100%
	SBU	1M	0.8M	80%
	COCO Caption	0.6M	0.6M	100%
Chinese	LAION-zh	108M	105M	97%
	In-house Data	220M	220M	100%
	Total	5B	1.4B	28%

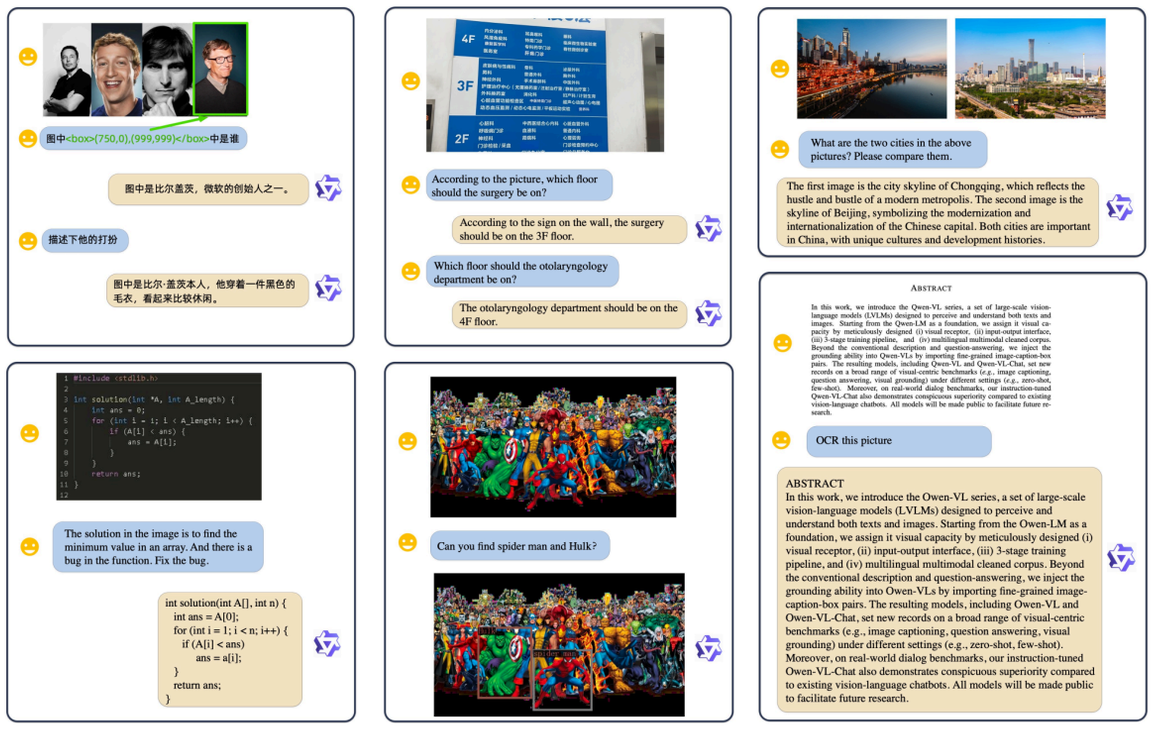
- Stage 2: higher quality task-specific data, increase resolution; train all parameters

> 阶段 2：更高质量的任务特定数据，提高分辨率；训练全部参数

Task	# Samples	Dataset
Captioning	19.7M	LAION-en & zh, DataComp, Coyo, CC12M & 3M, SBU, COCO, In-house Data
VQA	3.6M	GQA, VGQA, VQAv2, DVQA, OCR-VQA, DocVQA, TextVQA, ChartQA, AI2D
Grounding[2]	3.5M	GRIT
Ref Grounding	8.7M	GRIT, Visual Genome, RefCOCO, RefCOCO+, RefCOCOg
Grounded Cap.	8.7M	GRIT, Visual Genome, RefCOCO, RefCOCO+, RefCOCOg
OCR	24.8M	SynthDoG-en & zh, Common Crawl pdf & HTML
Pure-text Autoregression	7.8M	In-house Data

- Stage 3: instruction tuning data; freeze visual encoder, train adaptor + LM

> 阶段 3：指令微调数据；冻结视觉编码器，训练适配器 + LM

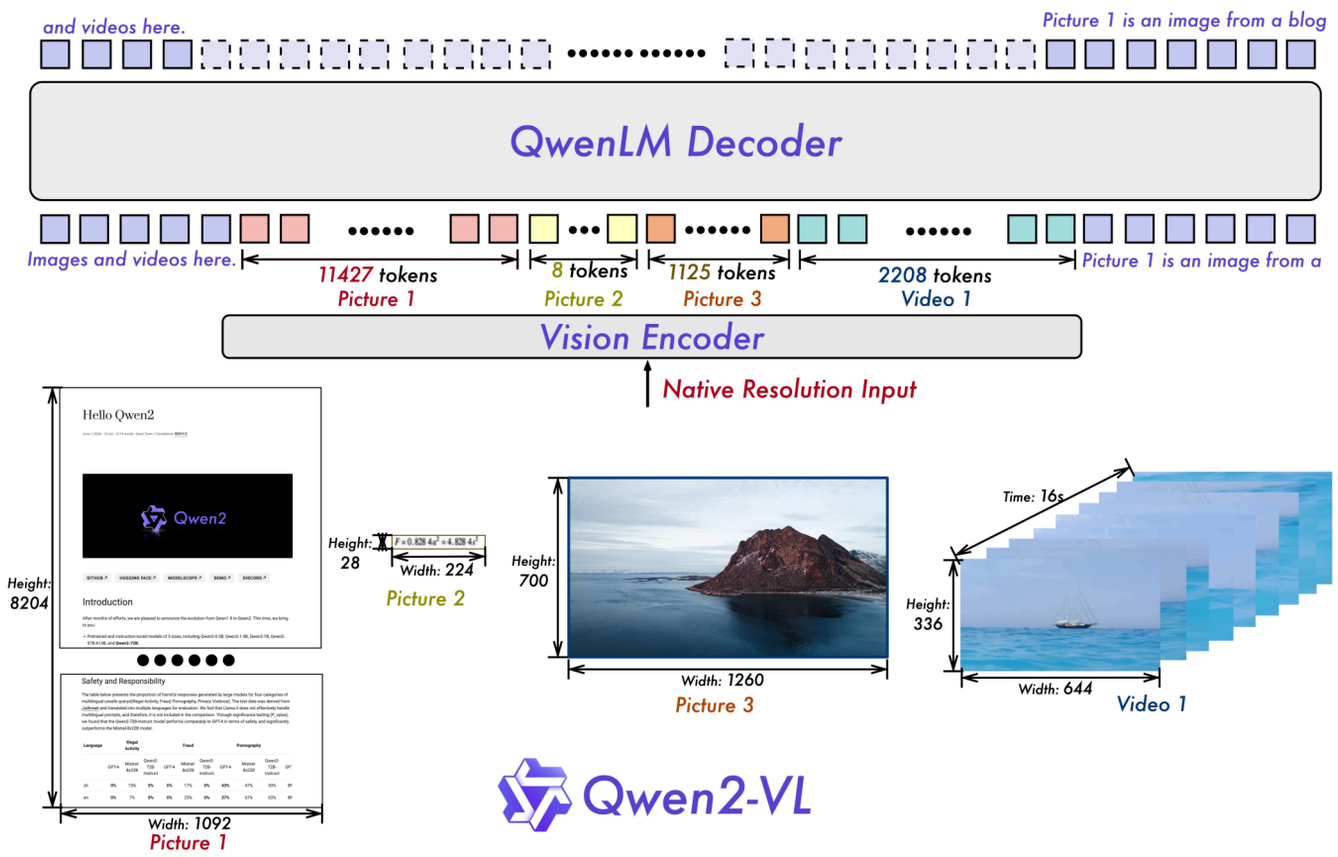


Qwen2-VL [Wang+ 2024]

> Qwen2-VL [Wang+ 2024]

Visual encoder: larger ViT (675M)

> 视觉编码器：更大的 ViT（675M）



- Key: dynamic resolution to handle varying resolutions

> 关键：用动态分辨率应对各种不同的分辨率

- Each 224 x 224 patch encoded with ViT/14, compress every 2x2 => 66 tokens

> 每个 224 x 224 patch 用 ViT/14 编码，每 2x2 压缩 => 66 个 token

- Video: sample 2 frames/sec, max 16384 tokens

> 视频：每秒采样 2 帧，最多 16384 个 token

Multimodal Rotary Position Embedding (MRoPE):

> 多模态旋转位置编码（MRoPE）：



Initialize LM with Qwen2 and vision encoder from DFN [Fang+ 2023]

> LM 用 Qwen2 初始化，视觉编码器来自 DFN [Fang+ 2023]

Training (similar to Qwen-VL):

> 训练（与 Qwen-VL 类似）：

- Stage 1: train only visual encoder

> 阶段 1：只训练视觉编码器

- Stage 2: train all parameters

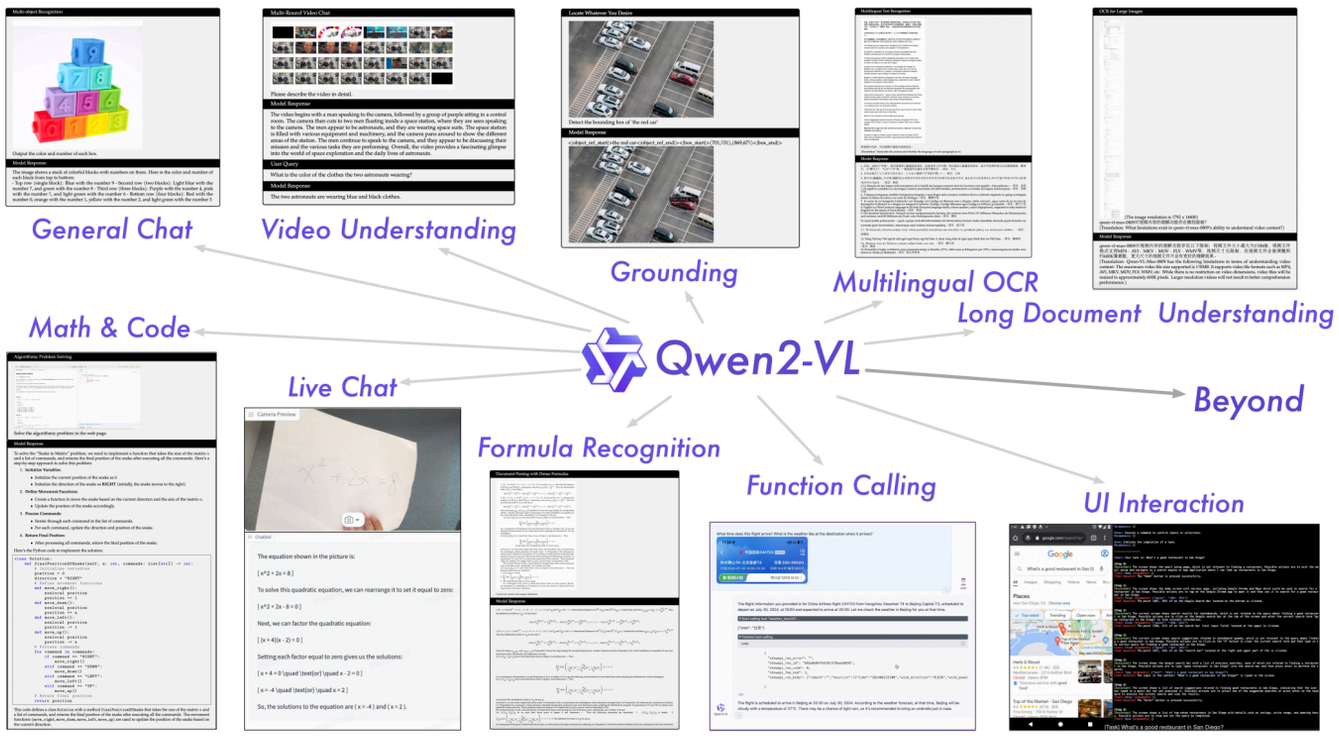
> 阶段 2：训练全部参数

- Stage 3: train language model on instruction following datasets

> 阶段 3：在指令遵循数据集上训练语言模型

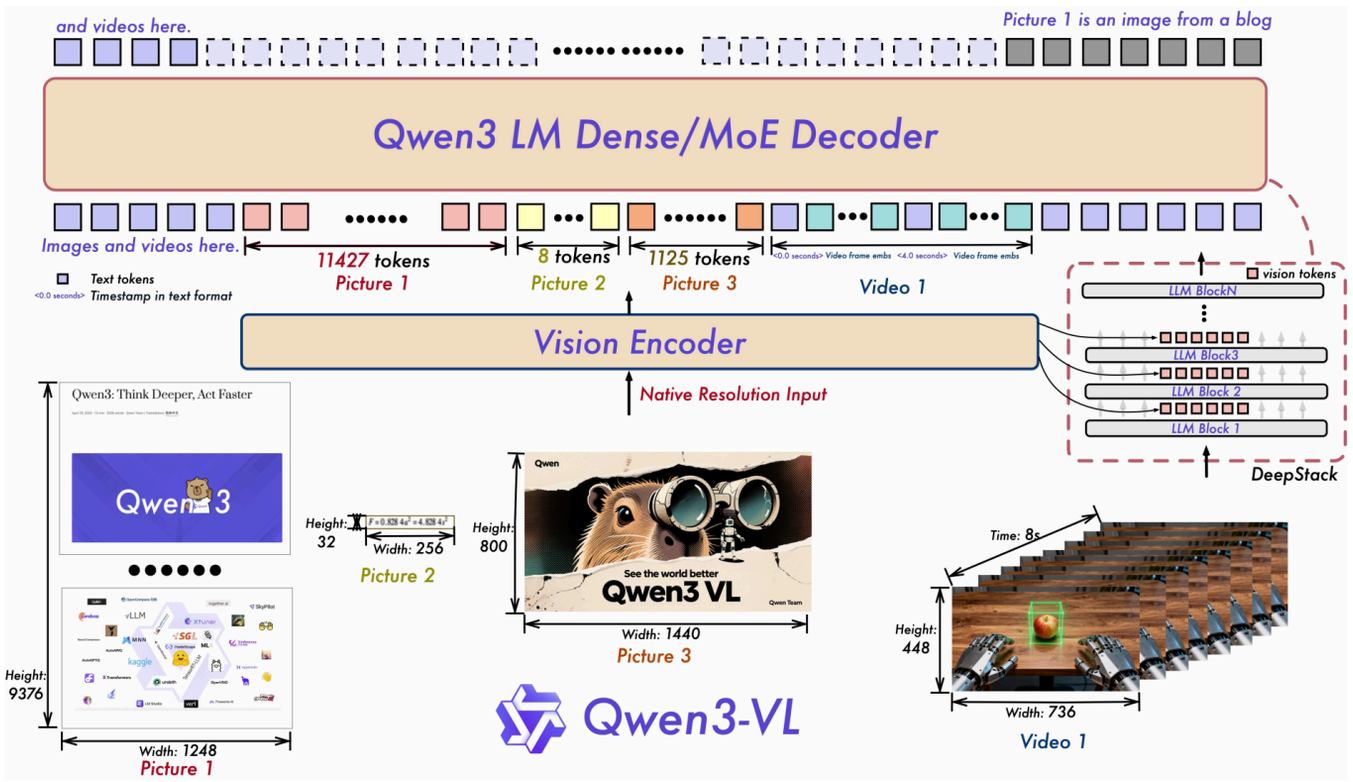
Many capabilities:

> 许多能力：



Qwen3-VL [Bai+ 2025]

> Qwen3-VL [Bai+ 2025]



Language model:

> 语言模型：

- Qwen-3 models (dense and MoE models up to 235B-A22B)

> Qwen-3 模型（dense 与 MoE 模型，最大到 235B-A22B）

- Long context understanding (256K)

> 长上下文理解（256K）

Vision encoder:

> 视觉编码器：

- SigLIP-2 (same architecture as SigLIP) [Tschannen+ 2025]

> SigLIP-2（与 SigLIP 架构相同） [Tschannen+ 2025]

- Interleaved MRoPE: distribute all axes (temporal, width, height) to low- and high-frequency bands

> Interleaved MRoPE：把所有轴（时间、宽、高）分配到低频与高频段

... [t w h t w h t w h t w h] rather than [t t t t w w w w h h h h]

> ... [t w h t w h t w h t w h] 而不是 [t t t t w w w w h h h h]

- Add explicit video timestamps (as separate tokens rather in positional embeddings)

> 加入显式的视频时间戳（作为独立的 token，而不是放进位置编码）

- Square-root-normalized per-token loss: balance text and multimodal data (video examples are long, don't want to dominate)

> 按平方根归一化的 per-token loss：平衡文本与多模态数据（视频样本很长，不想让它主导）

Adapter:

> 适配器：

- DeepStack: cross-layer fusion to inject visual information into multiple layers [Meng+ 2024]

> DeepStack：跨层融合，把视觉信息注入多个层 [Meng+ 2024]

Training:

> 训练：

- Pre-training has 4 stages (train adapter, train all parameters on 8K, 32K, 256K lengths)

> 预训练有 4 个阶段（训练 adapter；在 8K、32K、256K 长度上训练全部参数）

Stage	Objective	Training	Token Budget	Sequence Length
S0	Vision-Language Alignment	Merger	67B	8,192
S1	Multimodal Pre-Training	All	~1T	8,192
S2	Long-Context Pre-Training	All	~1T	32,768
S3	Ultra-Long-Context Adaptation	All	100B	262,144

- Post-training: SFT on long CoT data, knowledge distillation, RL

> 后训练：在长 CoT 数据上做 SFT、知识蒸馏、RL

	Benchmark	Qwen3-VL 235B-A22B		Gemini 2.5 Pro		OpenAI GPT-5		Claude Opus 4.1	
		thinking	instruct	thinking	budget-128	high	minimal	thinking	non-thinking
STEM Puzzle	MMMU	80.6	78.7	81.7*	80.9	**84.2***	74.4*	78.4	77.2
	MMMU-Pro	69.3	68.1	68.8*	71.2	**78.4***	62.7*	64.8	60.7
	$\text{MathVista}_{\text{mini}}$	**85.8**	84.9	82.7*	77.7	81.3	50.9	75.5	74.5
	MathVision	**74.6**	66.5	73.3*	66.0	70.9	45.8	64.3	57.7
	$\text{MathVision}_{\text{WP}}$	**63.8**	57.0	63.2	56.9	62.8	40.1	54.0	46.4
	We-Math	74.8	67.5	**80.6**	74.5	73.8	51.8	65.2	60.2
	$\text{MathVerse}_{\text{mini}}$	**85.0**	72.5	82.9	65.9	84.1	43.0	70.6	68.1
	DynaMath	82.8	79.4	80.0	78.5	**85.4**	74.0	75.1	72.0
	Math-VR	**66.8**	65.0	64.7*	54.3	58.1	21.7	54.3	38.0
	ZeroBench	**4**	2	3	1	2	2	3	1
	VlmsAreBlind	79.5	80.4	**86.1**	78.5	80.5	53.4	77.8	72.2
	LogicVista	**72.2**	65.8	72.0	68.7	71.8	46.3	67.3	63.5
	VisuLogic	**34.4**	29.9	31.6	26.9	28.5	27.2	27.9	27.2
	VisualPuzzles	57.2	54.7	**60.9**	56.9	57.3	47.9	48.8	47.6
General VQA	MMBench-EN	88.8	89.3	**90.1***	88.4	83.8	81.3	79.4	83.0
	MMBench-CN	88.6	88.9	**89.7***	86.4	83.5	79.9	84.9	74.3
	RealWorldQA	81.3	79.2	78.0*	76.0	**82.8**	77.3	69.9	68.5
	MMStar	**78.7**	78.4	77.5*	78.5	76.4	65.2	72.1	71.0
	SimpleVQA	61.3	63.0	**65.4**	66.9	61.8	56.7	56.7	55.7
Alignment	HallusionBench	**66.7**	63.2	63.7*	60.9	65.7	53.7	60.4	55.1
	MM-MT-Bench	**8.5**	8.5	8.4*	7.6	7.6	7.5	7.8	7.9
	MIA-Bench	**92.7**	91.3	92.3	91.3	92.4	92.6	91.2	90.0
Document Understanding	DocVQA_{test}	**96.5**	97.1	92.6	94.0	91.5	89.6	92.5	89.2
	InfoVQA_{test}	**89.5**	89.2	84.2	82.9	79.0	69.9	69.4	60.9
	$\text{AI2D}_{\text{w. M.}}$	89.2	89.7	**90.9**	90.0	89.7	84.1	86.4	84.4
	ChartQA_{test}	**90.3**	90.3	83.3	62.6	59.7	59.1	86.2	83.9
	OCRBench	**875**	920	866	872	810	787	764	750
	$\text{OCRBench_v2}_{\text{en}}$	**66.8**	67.1	54.3	55.2	53.0	48.2	48.4	47.2
	$\text{OCRBench_v2}_{\text{zh}}$	**63.5**	61.8	48.5	53.1	43.2	37.7	43.7	38.0
	CC-OCR	**81.5**	82.2	77.2	76.8	68.3	66.1	69.1	66.0
	$\text{OmniDocBench}_{\text{en}}$	**0.155**	0.143	0.347	0.206	0.356	0.174	0.194	-
	$\text{OmniDocBench}_{\text{zh}}$	**0.207**	0.207	0.238	0.249	0.472	0.389	0.293	-
	CharXiv(DQ)	90.5	89.4	**94.4**	87.8	89.2	79.5	88.5	87.8
	CharXiv(RQ)	66.1	62.1	67.9	62.9	**81.1***	57.8	63.6	60.2
	$\text{MMLongBench}_{\text{Doc}}$	**56.2**	57.0	55.6	51.2	51.5	42.4	54.5	48.1
2D/3D Grounding	RefCOCO-avg	**92.1**	91.9	74.6*	-	66.8	-	-	-
	CountBench	**93.7**	93.0	91.0*	91.0	91.7	87.8	93.1	91.9
	ODinW-13	**43.2**	48.6	33.7*	34.5	-	-	-	-
	ARKitScenes	**53.7**	56.9	-	-	-	-	-	-
	Hypersim	**11.0**	13.0	-	-	-	-	-	-
	SUNRGBD	**34.9**	39.4	29.7	-	-	-	-	-
Embodied/Spatial Understanding	ERQA	52.5	51.3	55.3	50.3	**65.7***	42.0*	34.8	28.0
	VSI-Bench	**60.0**	62.7	-	-	-	-	-	-
	EmbSpatialBench	**84.3**	83.1	79.1	73.3	82.9	75.1	69.2	66.0
	RefSpatialBench	**69.9**	65.5	36.5	35.6	23.8	23.1	-	-
	RoboSpatialHome	**73.9**	69.4	47.5	49.2	53.5	43.6	-	-
Multi-Image	BLINK	67.1	70.7	70.6*	70.0	**71.0**	62.8	64.1	62.9
	MUIRBENCH	**80.1**	73.0	77.2	74.0	77.5	66.5	-	-
Video Understanding	MVBench	75.2	76.5	69.9	65.8	**75.3**	64.6	61.4	59.0
	$\text{Video-MME}_{\text{w/o sub.}}$	79.0	79.2	**85.1**	80.6	84.7	77.3	75.6	73.3
	$\text{MLVU}_{\text{M-Avg}}$	83.8	84.3	85.6	81.2	**86.2**	78.3	73.5	71.2
	LVBench	63.6	67.7	**73.0**	69.0	-	-	-	-
	$\text{Charades-STA}_{\text{mIoU}}$	**63.5**	64.8	-	-	-	-	-	-
	VideoMMMU	80.0	74.7	83.6*	79.4	**84.6***	61.6*	76.2	70.1
	MMVU	71.1	68.1	**74.9**	72.2	73.0	68.1	66.4	61.4
Perception with Tool	V*	**85.9**	93.7+	83.8	72.7	72.8	56.7	-	-
	HRBench4K	84.3	85.4+	**87.3**	84.8	-	-	-	-
	HRBench8K	76.6	82.4+	**85.4**	80.1	-	-	-	-
Multi-Modal Coding	Design2Code	**93.4**	92.0	89.2	90.3	92.5	88.9	88.5	85.3
	ChartMimic	78.4	80.5	83.9	79.9	62.1	41.4	**85.2**	82.9
	UniSVG	65.8	69.8	70.0	67.9	71.7	74.5	**73.0**	72.5
Multi-Modal Agent	ScreenSpot Pro	**61.8**	62.0	-	-	-	-	-	-
	OSWorldG	**68.3**	66.7	45.2	-	-	-	-	-
	AndroidWorld	**62.0**	63.7	-	-	-	-	-	-
	OSWorld	**38.1**	31.6	-	-	-	-	-	44.4
	WindowsAA	**32.1**	28.9	-	-	-	-	-	-

Summary:

> 总结：

- SOTA performance

> SOTA 性能

- Lots of data work, but not many details

> 数据工作上投入很大，但细节不多

- Minor but potentially important architectural improvements

> 一些次要但可能重要的架构改进

- Scale up

> 扩大规模

Chameleon [Chameleon Team 2024]

> Chameleon [Chameleon Team 2024]

So far: VLMs encode images (via CLIP or SigLIP), inject into LM

> 目前为止：VLM（通过 CLIP 或 SigLIP）编码图像并注入 LM

Disadvantage: can't generate images (need diffusion)

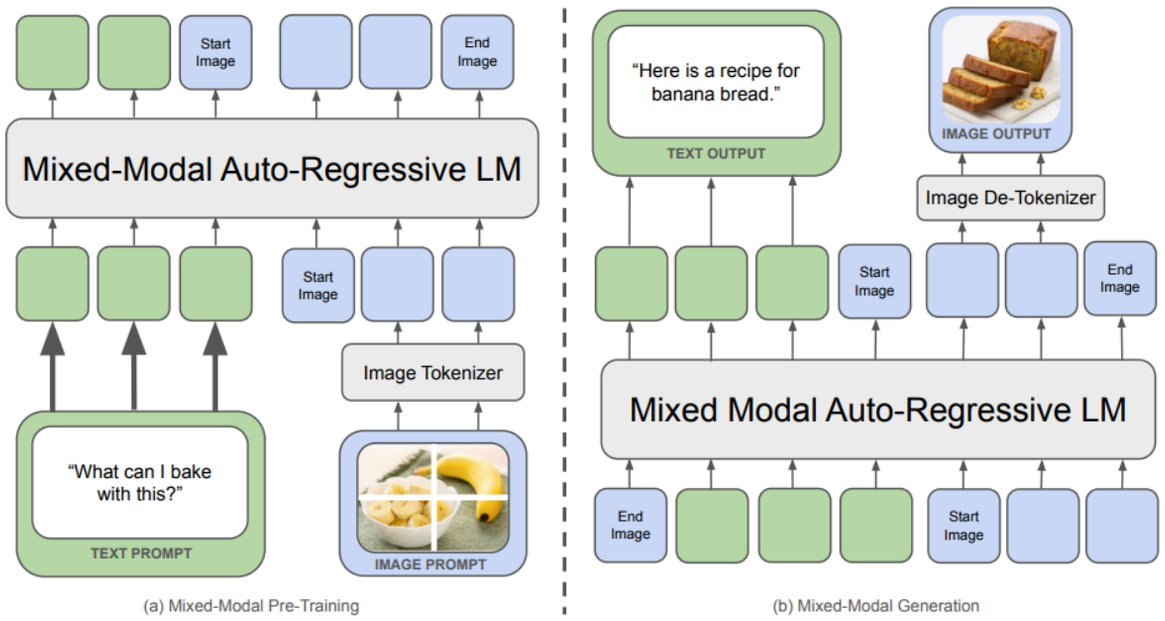
> 缺点：无法生成图像（需要 diffusion）

Chameleon: map everything into discrete tokens

> Chameleon：把一切都映射成离散 token

Advantage: can analyze and generate images in a uniform way

> 优点：能以统一的方式分析和生成图像



Prompt: I'm bored. Could you show me some cool, quirky-looking birds? Also tell me about them in 2–3 sentences.

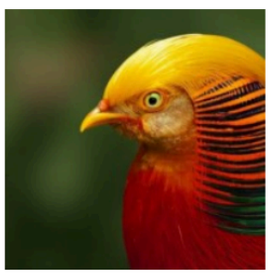


Response: Absolutely! Here are three birds with some unique features that you might find interesting:

Keel-Billed Toucan The Keel-Billed Toucan is a brightly colored bird native to South American rainforests and woodlands. Its most distinctive feature is its colorful beak, which can grow up to seven inches long and has a distinctive, curved shape. The beak is used for plucking fruit from trees and for social displays. Despite its large size, the Keel-Billed Toucan is a skilled flyer and can often be seen flitting through the treetops. Here is an image of this vibrant bird: **<img>**



Puffin Puffins are small, aquatic birds that are found in the northern hemisphere. They have distinctive black and white feathers, as well as a brightly colored beak that is used for attracting mates and defending their territory. Puffins are known for their distinctive "puffin" call, which sounds like a loud, guttural chuckle. They feed on fish, which they catch by diving into the water from the air. Puffins are colonial birds, and they breed in large colonies on rocky cliffs and islands. Here is an image of this adorable bird: **<img>**



Golden Pheasant The Golden Pheasant is a brightly colored bird native to the forests of China. It is a member of the pheasant family and is known for its striking plumage, which features a golden-yellow body, red face, and green tail feathers. The Golden Pheasant is a ground-dwelling bird, and it feeds on a diet of insects, seeds, and fruits. It is a popular game bird, and is often hunted for its meat and feathers. In captivity, the Golden Pheasant is sometimes kept as an ornamental bird, due to its striking appearance. Here is an image of this beautiful bird: **<img>**

I hope these images and descriptions have helped to brighten your day! If you have any other questions or requests, please don't hesitate to ask.

Vision encoder [Gafni+ 2022]

> 视觉编码器 [Gafni+ 2022]

- Key difference: encoder needs to map to discrete tokens (so we can generate them)

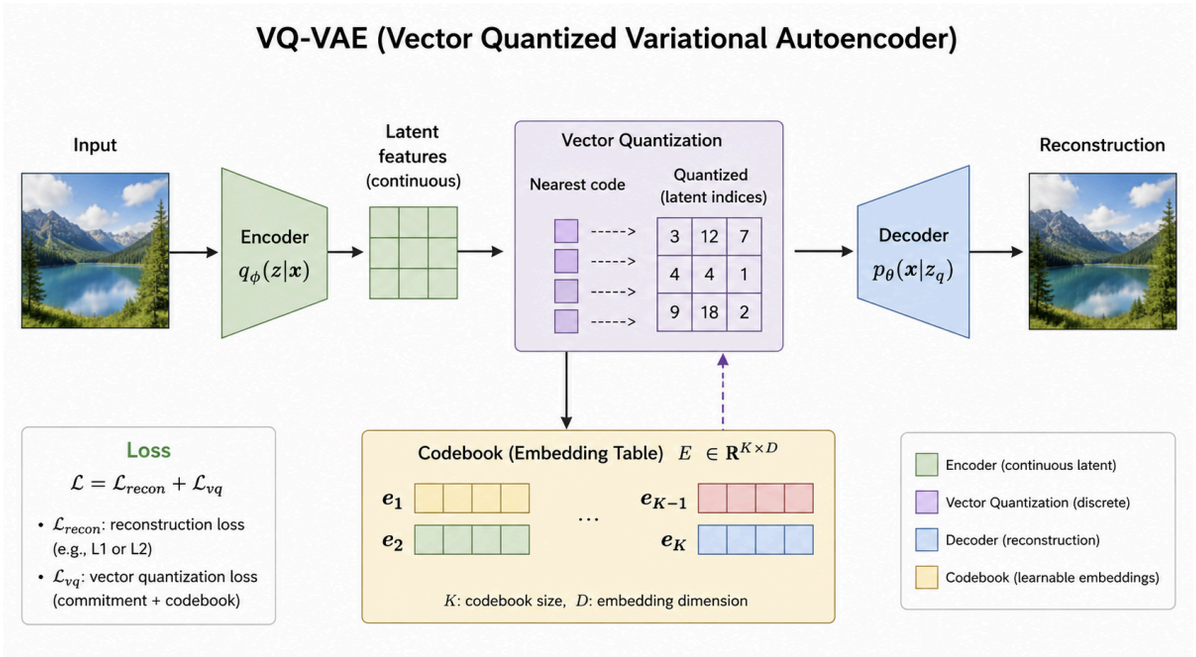
> 关键区别：编码器需要映射到离散 token（这样我们才能生成它们）

- VQ-VAE (Vector Quantized Variational Autoencoder) [Oord+ 2017]

> VQ-VAE（向量量化变分自编码器） [Oord+ 2017]

- Idea: map image to a discrete codebook, decode back to image and minimize reconstruction loss

> 思路：把图像映射到离散码本，再解码回图像，并最小化重构损失



- Encodes 512 x 512 image into 1024 tokens (codebook of size 8192)

> 把 512 x 512 的图像编码成 1024 个 token（码本大小 8192）

- Train a new BPE tokenizer

> 训练一个新的 BPE tokenizer

Training:

> 训练：

- Stage 1 (80%): large-scale, unsupervised (2.9T text tokens, 1.5T text/image tokens, 400B text/image interleaved tokens)

> 阶段 1（80%）：大规模、无监督（2.9T 文本 token，1.5T 文本/图像 token，400B 图文交错 token）

- Stage 2 (20%): 50% of stage 1 data, 50% of high quality data

> 阶段 2（20%）：50% 用阶段 1 的数据，50% 用高质量数据

Training stability

> 训练稳定性

- Text tokens have low entropy, image tokens have high entropy, leads to norm growth, logit drift problem

> 文本 token 熵低，图像 token 熵高，会导致范数增长、logit 漂移问题

- Fixes: QK norm, z-loss regularization

> 修复手段：QK norm、z-loss 正则化

Summary:

> 总结：

- Elegant (just autoregressive modeling of discrete tokens)

> 优雅（只是对离散 token 做自回归建模）

- Not as performant (discretization loses information - think OCR)

> 性能没那么好（离散化会损失信息——想想 OCR）

- Training with multiple modalities is tricky

> 多模态一起训练并不容易